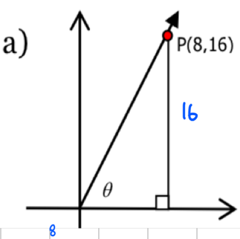
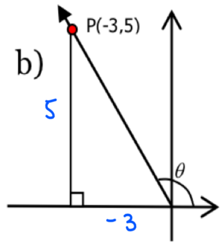
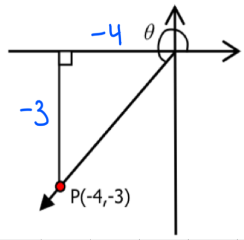
3.2 Practice SOLUTIONS

1. The co-ordinates of point P on the terminal arm of each $\angle\theta$ in standard position are shown,
where $0 \le \theta \le 2\pi$. Determine the exact values of $\sin(\theta)$, $\cos(\theta)$, and $\tan(\theta)$.

a)


b)


c)


a) $r = \sqrt{x^2+y^2} = \sqrt{16^2+8^2} = \sqrt{320} = 8\sqrt{5}$

$\therefore \sin\theta = \frac{16}{8\sqrt{5}} = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$

$\cos\theta = \frac{8}{8\sqrt{5}} = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}$

$\tan\theta = \frac{16}{8} = 2$

b) $r = \sqrt{x^2+y^2} = \sqrt{(-3)^2+5^2} = \sqrt{34}$

$\therefore \sin\theta = \frac{5}{\sqrt{34}}$

$\cos\theta = -\frac{3}{\sqrt{34}} = -\frac{3\sqrt{34}}{34}$

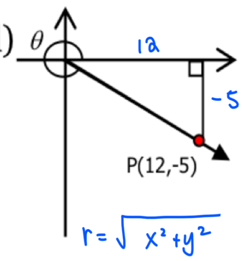
$\tan\theta = -\frac{5}{3}$

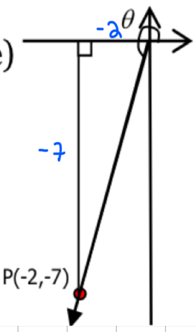
c) $r = \sqrt{x^2+y^2} = \sqrt{16+9} = 5$

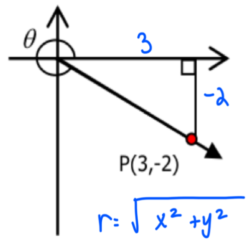
$\therefore \sin\theta = -\frac{3}{5}$

$\cos\theta = -\frac{4}{5}$

$\tan\theta = 3/4$

d)


e)


f)


d) $r = \sqrt{x^2+y^2} = \sqrt{25+144} = 13$

$\therefore \sin\theta = -\frac{5}{13}$

$\cos\theta = \frac{12}{13}$

$\tan\theta = -\frac{5}{12}$

e) $r = \sqrt{x^2+y^2} = \sqrt{(-7)^2+(-2)^2} = \sqrt{53}$

$\therefore \sin\theta = -7/\sqrt{53}$

$\cos\theta = -2/\sqrt{53}$

$\tan\theta = 7/2$

f) $r = \sqrt{x^2+y^2} = \sqrt{3^2+(-2)^2} = \sqrt{13}$

$\therefore \sin\theta = -2/\sqrt{13}$

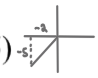
$\cos\theta = 3/\sqrt{13}$

$\tan\theta = -2/3$

2. The co-ordinates of point P on a terminal arm of each $\angle\theta$ in standard position are shown, where $0 \le \theta \le 2\pi$. Determine the exact values of $\sin(\theta)$, $\cos(\theta)$, and $\tan(\theta)$.

a) $P(6,5)$


b) $P(-1,8)$


c) $P(-2,-5)$


d) $P(6,-1)$


a) $r = \sqrt{61}$

$\therefore \sin\theta = 5/\sqrt{61}$

$\cos\theta = 6/\sqrt{61}$

$\tan\theta = 5/6$

b) $r = \sqrt{65}$

$\therefore \sin\theta = 8/\sqrt{65}$

$\cos\theta = -1/\sqrt{65}$

$\tan\theta = -8$

c) $r = \sqrt{29}$

$\therefore \sin\theta = -5/\sqrt{29}$

$\cos\theta = -2/\sqrt{29}$

$\tan\theta = 5/2$

d) $r = \sqrt{37}$

$\therefore \sin\theta = -1/\sqrt{37}$

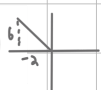
$\cos\theta = 6/\sqrt{37}$

$\tan\theta = -1/6$

e) $P(2,-4)$


f) $P(-3,-9)$


g) $P(3,3)$


h) $P(-2,6)$


e) $r = \sqrt{20} = 2\sqrt{5}$

$\therefore \sin\theta = -4/2\sqrt{5} = -\frac{2}{\sqrt{5}} = \frac{-2\sqrt{5}}{5}$

$\cos\theta = \frac{2}{2\sqrt{5}} = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}$

$\tan\theta = -4/2 = -2$

f) $r = \sqrt{90} = 3\sqrt{10}$

$\therefore \sin\theta = \frac{-9}{3\sqrt{10}} = \frac{-3}{\sqrt{10}} = \frac{-3\sqrt{10}}{10}$

$\cos\theta = \frac{-3}{3\sqrt{10}} = \frac{-1}{\sqrt{10}} = \frac{-\sqrt{10}}{10}$

$\tan\theta = -9/-3 = 3$

g) $r = \sqrt{18} = 3\sqrt{2}$

$\therefore \sin\theta = \frac{3}{3\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

$\cos\theta = \frac{3}{3\sqrt{2}} = \frac{\sqrt{2}}{2}$

$\tan\theta = 1$

h) $r = \sqrt{40} = 2\sqrt{10}$

$\therefore \sin\theta = \frac{6}{2\sqrt{10}} = \frac{3}{\sqrt{10}} = \frac{3\sqrt{10}}{10}$

$\cos\theta = \frac{-2}{2\sqrt{10}} = \frac{-1}{\sqrt{10}} = \frac{-\sqrt{10}}{10}$

$\tan\theta = \frac{6}{-2} = -3$

3. Find the exact value of each trigonometric ratio.

a) $\sin\left(\frac{5\pi}{4}\right)$ b) $\tan\left(\frac{11\pi}{6}\right)$ c) $\cos\left(\frac{\pi}{6}\right)$ d) $\cos\left(\frac{7\pi}{4}\right)$

a) $= \sin\left(\pi + \frac{\pi}{4}\right)$
$= -\sin\left(\frac{\pi}{4}\right)$
$= -\frac{1}{\sqrt{2}}$
$= -\sqrt{2}/2$

b) $= \tan\left(2\pi - \frac{\pi}{6}\right)$
$= -\tan\left(\frac{\pi}{6}\right)$
$= -\frac{1}{\sqrt{3}}$
$= -\sqrt{3}/3$

c) $= \sqrt{3}/2$

d) $= \cos\left(2\pi - \frac{\pi}{4}\right)$
$= \cos\left(\frac{\pi}{4}\right)$
$= 1/\sqrt{2}$
$= \sqrt{2}/2$

e) $\tan\left(\frac{4\pi}{3}\right)$ f) $\cos\left(\frac{7\pi}{6}\right)$ g) $\sin\left(\frac{5\pi}{6}\right)$ h) $\cos\left(\frac{3\pi}{4}\right)$

e) $= \tan\left(\pi + \frac{\pi}{3}\right)$
$= \tan\left(\pi/3\right)$
$= \sqrt{3}$

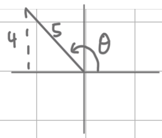
f) $= \cos\left(\pi + \pi/6\right)$
$= -\cos\left(\pi/6\right)$
$= -\sqrt{3}/2$

g) $= \sin\left(\pi - \pi/6\right)$
$= \sin\left(\pi/6\right)$
$= 1/2$

h) $= \cos\left(\pi - \pi/4\right)$
$= -\cos\left(\pi/4\right)$
$= -\frac{1}{\sqrt{2}}$
$= -\frac{\sqrt{2}}{2}$

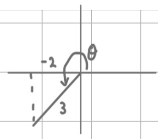
4. $\angle\theta$ in standard position with its terminal arm in the stated quadrant, and $0 \le \theta \le 2\pi$.
A trigonometric ratio is given. Find the exact values of the other two trigonometric ratios.

a) $\sin(\theta) = \frac{4}{5}$, quadrant II b) $\cos(\theta) = -\frac{2}{3}$, quadrant III

a)


$r^2 = x^2 + y^2$
$5^2 = x^2 + 4^2$
$25 - 16 = x^2$
$9 = x^2$
$-3 = x$

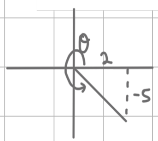
$\therefore \cos\theta = -\frac{3}{5}$
$\tan\theta = -4/3$

b)


$r^2 = x^2 + y^2$
$3^2 = (-2)^2 + y^2$
$9 - 4 = y^2$
$-\sqrt{5} = y$

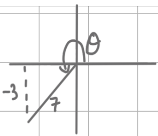
$\therefore \sin\theta = \frac{-\sqrt{5}}{3}$
$\tan\theta = \frac{\sqrt{5}}{2}$

c) $\tan(\theta) = -\frac{5}{2}$, quadrant IV d) $\sin(\theta) = -\frac{3}{7}$, quadrant III

c)


$r^2 = x^2 + y^2$
$r^2 = 2^2 + (-5)^2$
$r^2 = 4 + 25$
$r = \sqrt{29}$

$\therefore \cos\theta = \frac{2}{\sqrt{29}} = \frac{2\sqrt{29}}{29}$
$\sin\theta = \frac{-5}{\sqrt{29}} = \frac{-5\sqrt{29}}{29}$

d)


$r^2 = x^2 + y^2$
$7^2 = x^2 + (-3)^2$
$49 - 9 = x^2$
$-\sqrt{40} = x$
$-2\sqrt{10} = x$

$\therefore \cos\theta = \frac{-2\sqrt{10}}{7}$
$\tan\theta = \frac{3}{2\sqrt{10}} = \frac{3\sqrt{10}}{20}$

5. Determine $\angle\theta$, given $0 \le \theta \le 2\pi$.

a) $\sin(\theta) = \frac{1}{2}$ b) $\cos(\theta) = \frac{\sqrt{3}}{2}$ c) $\tan(\theta) = -1$ d) $\sec(\theta) = -2$

a) R.A.A $= \pi/6$
$\theta_1 = \frac{\pi}{6}$
$\theta_2 = \pi - \frac{\pi}{6} = 5\pi/6$

b) R.A.A $= \pi/6$
$\theta_1 = \pi/6$
$\theta_2 = 2\pi - \pi/6 = \frac{5\pi}{3}$

c) R.A.A $= \frac{\pi}{4}$
$\theta_1 = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$
$\theta_2 = 2\pi - \frac{\pi}{4} = 7\pi/4$

d) $\frac{1}{\cos\theta} = -2$
$\cos\theta = -\frac{1}{2}$
R.A.A $= \pi/3$
$\theta_1 = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$
$\theta_2 = \pi + \frac{\pi}{3} = 4\pi/3$

e) $\cot(\theta) = -\sqrt{3}$ f) $\sin(\theta) = -\frac{5}{6}$

e) $\frac{1}{\tan\theta} = -\sqrt{3}$
$\tan\theta = -\frac{1}{\sqrt{3}}$
R.A.A $= \pi/6$
$\theta_1 = \pi - \frac{\pi}{6} = 5\pi/6$
$\theta_2 = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$

f) R.A.A $= \sin^{-1}\left(\frac{5}{6}\right) \doteq 0.985$ rad
$\theta_1 = \pi + 0.985 = 4.127$ rad
$\theta_2 = 2\pi - 0.985 = 5.298$ rad